CHECKLIST OF COMPLIANCE WITH THE SYLLABUS PARTS AND FORMAT

Course Code and Title**: S-CSIS311 – INTRODUCTION TO MACHINE LEARNING LABORATORY**

This undertaking has a two-fold objective. First, it allows the syllabus designers to review their output systematically. Based on their honest evaluation, they could make the necessary revision. Second, it expedites the checking to be done by the Dept. Syllabus Committee Reviewer, Department Chair, the College Dean, and the Institutional Syllabus Reviewers.

DIRECTIONS: Review the latest version of the syllabus that you submitted to the Department Chair for his/her scrutiny. As you do your reflective review, As you do your reflective review, assess your compliance with the syllabus parts and format by answering each of the items that follow. Shade the corresponding box (complied with, not complied with) under the compliance column.

Format, parts, and description	Compliance	
	Complied With	**Not Complied With**
Format		
1. The 2023-2024 syllabus format is used.	/	
2. The font used in the syllabus preparation is Times New Roman (11).	/	
3. DLSUD (being first) and college logos are beside each other at the top (center).	/	
4. Preliminaries include all the contact details of the professor, such as the day and time.	/	
5. Each WEEK has a corresponding date.	/	
Content		
6. The Learning Outcomes have more HOTS than LOTS.	/	
7. CLOs and TLOs are identified for each module/lesson/topic.	/	
8. Gospel reading and Reflection and reflection precede the presentation of each topic under the teaching-learning activities.	/	
9. The journal articles/references are within the last five years.	/	
10. Some materials are beyond the five-year rule because these are considered cannon and seminal studies.	/	
11. The course policies and requirements section of the syllabus is a clear provision for addressing the concerns and grievances of students.	/	
Assessments		
12. At least three and a maximum of five Enabling Assessments per term are reflected in the syllabus.	/	
13. Formative Assessments are not reflected anymore in the syllabus.	/	
14. Enabling Assessments are briefly described and aligned with the learning outcomes and teaching-learning activities.	/	
15. The syllabus presents one-module(lesson)-one-assessment correspondence.	/	
16. Only one Enabling Assessment is reflected in a given week	/	
17. The Enabling Assessments are aligned with the identified CLOs and TLOs.	/	

18. Enabling Assessments, Class Participation, and Summative Assessments constitute the breakdown of grades per term.	/	
19. Thirty percent (30%) is allotted for Summative Assessments per term.	/	
20. Twenty percent (20%) is allotted for Class Participation/Attendance per term.	/	
21. The percentage that is allocated for Enabling Assessments fairly corresponds to their quantity, depth, breadth, and level of difficulty compared to the other components of the Midterm Grade and Final Grade.	/	
22. Summative Assessments are aligned with the identified CLOs, TLOs and teachinglearning activities.	/	
23. Enabling Assessments are aligned with the identified CLOs and TLOs and teachinglearning activities.	/	
24. The Midterm Summative Assessment measures appropriately and validly with the identified CLOs and TLOs for the entire MIDTERM PERIOD	/	
25. The Final Summative Assessment measures appropriately and validly with the identified CLOs and TLOs for the entire FINAL PERIOD.	/	
References and Curriculum Map		
26. The References must be within the last 5 years except for canons and seminal studies.	/	
27. It must be a combination of books, eBooks, journal articles, videos, etc.	/	
28. It should follow the APA 7th edition for CLAC, CBAA, CTHM, CCJE, CEAT, COED, and CSE for CSCS.	/	
29. The syllabus is submitted with an attached mapping table (CLOs with PLOs and ILOs)	/	

NOTE: Enabling Assessments include Discussion Type (cf. SB), Forum, Analysis Paper, Reports, etc.

Self-reviewed:

Rolando B. Barrameda

Checked, verified, and endorsed:

Rolando B. Barrameda
Dept. Curriculum Committee Reviewer

Checked, verified, and approved:

Josephine T. Eduardo
Department Chair

Dr. Rubie M. Causaren
College Dean

De La Salle University - Dasmariñas
COLLEGE OF SCIENCE AND COMPUTER STUDIES
COMPUTER STUDIES DEPARTMENT

COURSE SYLLABUS

COURSE CODE	**:**	S-CSIS311
COURSE TITLE	**:**	Introduction to Machine Learning
COURSE TYPE	**:**	Laboratory
LEARNING MODALITY:		Traditional Blended Learning
COURSE CREDIT	**:**	1 unit
PRE-REQUISITES	**:**	Should be 3rd year Level
CO-REQUISITES	**:**	None
INSTRUCTOR	**:**	ROLANDO BARRAMEDA
CONSULTATION	**:**	2:00 – 3:00 Monday
		rbbarrameda@dlsud.edu.ph
		(046)4811900 local 3134

COURSE DESCRIPTION:

This course deals with the fundamentals machine learning techniques, as well as hands-on experience applying these techniques and developing new techniques for solving problems from the real world; topics include regression (least square regression, lasso), classification (naive Bayes, nearest support vector machines, logistic regression), kernel methods, unsupervised methods (k-means clustering, dimensionality reduction), stochastic optimization, deep learning and recent advances in big data analytics. Student can attend seminars, workshops, and training in relevant disciplines onsite or online.

COURSE LEARNING OUTCOMES:

By the end of this course, students can:

CLO1. determine a wide variety of learning algorithms;
CLO2. demonstrate how to evaluate models generated from data;
CLO3. develop an appreciation for what is involved in learning models from data;

CLO4. apply the algorithm to a real-world problem; and
CLO5. formulate the optimize models learned and report on the expected accuracy that can be achieved by applying the models.

LEARNING PLAN :

Learning Outcomes for Midterm Period

Course Learning Outcomes	Topic Learning Outcomes
CLO1. Describe the principles of machine learning in various domains using applied learning algorithms.	• TLO1. Explain the machine learning in a nutshell. • TLO2. Demonstrate your knowledge of various algorithms; and • TLO3. Show your appreciation for the algorithms used in machine learning
CLO2. Demonstrate how to evaluate data driven models.	• TLO4. Identify and establish corpus data models. • TLO 5. Apply data pre-processing techniques to integrate essential elements. • TLO6. Apply working with datasets, splitting and feature calling.
CLO3. Develop an appreciation for what is involved in learning models from data.	• TL06. Support observation to best fitting the data. • TLO7. Classify domain areas of machine learning
CL04. Apply the algorithm to a real-world problem.	• TL08. Exhibits and demonstrate data observation. • TLO9. Generate and evaluate data observation using python.
CLO5. Formulate the optimize models learned and report on the expected accuracy that can be achieved by applying the models	• TLO10.Collaboration, individual assessment to design and create a research paper project.

Module No and Learning Outcome(s)	Week	Teaching-Learning Activities / Assessment Strategies Online Module Activities	Technology Enabler	Onsite	Summative	Allotted Hours

Module 1 TLO 1 TLO 2 TLO 3 TLO 4 CLO 1 CLO 2 CLO 5	1 – 2 8/14-8/26	• Gospel reading and reflection • Syllabus uploaded • Presentation of course introduction • Course requirement details	Google Colab Anaconda Schoolbook MS Teams	/		2.0
		• Gospel reading and reflection • Enabling Assessment: Matplot and Seaborn, Data Types, Mean, Median, Mode, SD and Variance	Google Colab Anaconda Schoolbook MS Teams	/		2.0
Module 2 TLO 3 TLO 4 CLO 2 CLO 5	3 8/28-9/2	• Gospel reading and reflection • Enabling Assessment: Conditional Probability, Linear, Polynomial Regression, Multiple Regression	Google Colab Anaconda Schoolbook MS Teams	/		2.0
Module 3 TLO 5 TLO 6 CLO 3 CLO 5	4 9/4-9/9	• Gospel reading and reflection • Enabling assessment : K-Nearest Neighbors	Google Colab Anaconda Schoolbook MS Teams	/		2.0
	5 9/11-9/16			/		2.0
	6 9/18-9/23			/		2.0
	7 9/25-9/30	• Gospel reading and reflection • Enabling assessment : K-Fold Cross Validation Split on Train and Test Set	Google Colab Anaconda Schoolbook MS Teams	/		2.0
	8 10/2-10/7	• Gospel reading and reflection • Enabling assessment : Support Vector Machine	Google Colab Anaconda Schoolbook MS Teams	/		2.0
	9 10/9-10/14	• SUMMATIVE ASSESSMENT (Module 1 – 4)	Schoolbook		/	2.0
			TOTAL	***16.0***	***2.0***	***18.0***

Learning Outcomes for Final Period

Course Learning Outcomes	Topic Learning Outcomes
CLO3. Develop an appreciation for what is involved in learning models from data.	• TL06. Support observation to best fitting the data • TLO7. Classify domain areas of machine learning
CL04. Apply the algorithm to a real-world problem and other domain areas.	• TL08. Exhibits and demonstrate data observation.

	• TLO9. Generate and evaluate data observation using python.

Module No and Learning Outcome(s)	Week	Teaching-Learning Activities / Assessment Strategies	Technology Enabler	Onsite	Self Care	Summative	Allotted Hours
		Online Module Activities					
Module 4 TLO 5 TLO 6 CLO 3 CLO 5	10 10/16-10/21	• Gospel reading and reflection • Enabling Assessment: Naïve Bayes and K-Means	Google Colab Anaconda Schoolbook MS Teams	/			2.0
	11 10/23-10/28	SELF-CARE / ME-TIME			/		2.0
Module 5 TLO 5 TLO 6 CLO 3 CLO 5	12 10/30-11/4	• Gospel reading and reflection • Research Collaboration • Enabling assessment: Chapter 1 and 2	Schoolbook MS Teams	/			2.0
	13-14 11/6-11/18	• Gospel reading and reflection • Research Collaboration • Enabling assessment: Chapter 3 and 4	Schoolbook MS Teams	/			4.0
Module 6 TLO 5 TLO 6 CLO 3 CLO 5	15 11/20-11/25	Gospel reading and reflection Research Collaboration Enabling assessment: Chapter 5,6 and 7	Schoolbook MS Teams	/			2.0
	16 11/27-12/2	• Forum, Collaboration, Research Presentation • Enabling assessment: Paper Submission Conference (Local / International)	Schoolbook MS Teams	/			2.0
	17 12/4-12/9	• Completion of Final Output	Schoolbook	/			2.0
	18 12/11-12/16	• SUMMATIVE ASSESSMENT (Module 5 – 7)	Schoolbook			/	2.0

TOTAL	***14.0***	***2.0***	***2.0***	***18.0***

GRADING SYSTEM:

Midterm

Attendance/Class Participation	- 20%
Enabling Assessments	- 50%
Summative Assessments	- 30%
	100%

*Seminar, training and workshop online /onsite consider as part of enabling and class participation

Finals

Attendance/Class Participation	- 20%
Enabling Assessments	- 50%
Summative Assessments	- 30%
	100%

Overall Final Grade = $\frac{\text{Midterm + Final}}{2}$

COURSE POLICIES AND REQUIREMENTS:

1. This course follows the Traditional Blended Learning Mode. There will be 50% onsite and 50% asynchronous.

2. Enrollment in an E-Class. The Center for Innovative Learning Programs (CILP) shall enroll you in the e-class. This will be based on the class list provided by the Office of the University Registrar and at the portal.

3. Communication and Feedback
    a. Onsite classes shall happen as indicated in the learning plan section of the syllabus. Fifty percent (50%) of the weekly classes are delivered face-to-face while the remaining is asynchronous.
    b. Online Asynchronous Communication shall happen when the faculty utilizes at least one of the recommended asynchronous communications. The minimum response time for asynchronous communication is 48 hours, excluding Sundays and holidays. All communication platforms must be explicitly indicated in the syllabus. Below are the recommended asynchronous communication platforms:
        i. Forum – at the Forum Tab at Schoolbook, students can post their general concerns. The forum is public, and any post can be seen by the class.
        ii. Messaging – at Schoolbook. Any private or confidential matters may be messaged through Schoolbook messaging feature.
        iii. Email – you may email your professor. (Professor's email will be given during the course orientation)
    c. No use of foul, curse, and other bad words, whether written or verbal. No shouting. No flooding of chat and messages. No selling.

4. Attendance Policy
    a. Attendance Policy for Onsite Classes is scheduled based on the learning plan. A maximum of 20% for class participation/recitation/attendance shall be allotted for the student's grade.
    b. The onsite meeting will only be held at the schedule indicated in the registration form.
    c. Students are required to attend onsite classes, 40% of absence (from onsite classes) for the entire semester means 0.0 final grade.

5. Intellectual Property Policy
    a. Policy and copyright infringement will not be tolerated. Any violation of such will earn students a grade of zero in the assessment.
    b. Using any illegally obtained software is strictly prohibited.
    c. Students are encouraged to use their original photos, videos, and other resources. Otherwise, students can use royalty-free resources or embed the sources in their submissions to avoid copyright infringement and/or plagiarism.
    d. Giving of password to Schoolbook and Office 365 is strictly prohibited. Likewise, accessing Schoolbook and Office 365 account other than the students' personal account is also strictly prohibited. Violating students will be reported to the Student Welfare and Formation Office (SWAFO).

6. Other Policies
    a. Attire - students must wear decent casual or smart-casual attire prescribed by the Student Handbook.
    b. Students must be always honest; cheating and plagiarism in any form will merit a grade of 0.00.
    c. Cellular/Mobile phones should always be in silent mode during class hours; the use of cellular phones is prohibited in class unless special permission is sought. Cellular phones cannot also be used as calculator during examination.
    d. Any complaints against the teacher or against classmates should be properly addressed to the subject teacher for appropriate action. Students may seek the help and guidance of their academic/registration adviser in resolving the issue with the subject teacher when such mediation is needed.

REFERENCES :

On-Site Book References

Call number or e-provider	**Reference Material**
QA 76.73.P98 .L510 2019	Wei-Meng Lee. 2019. Python machine learning. John Wiley & Sons, Inc.
TK 5102.9 .L721 2019	Max Little. 2019. Machine learning for signal processing: data science, algorithms, and computational statistics. Oxford University Press.
HG 4521 .G946 2019eb	Tony Guida. 2019. Big data and machine learning in quantitative investment. John Wiley & Sons, Inc.
Q 325.7 .C382 2018	Eugener Charniak. 2018. Introduction to Deep Learning. The MIT Press.
QA 76.9.B45 .K28 2018	John Kelleher D. and Brendan Tierney. 2018. Data science. The MIT Press
DLSU-D E- Books References:	

Accession Number: 3501568	Katharina Morik, Jörg Rahnenführer, Christian Wietfeld. 2023. Machine Learning Under Resource Constraints – Applications. Volume 3, Applications.
Accession Number: 1881497	Giuseppe Bonaccorso. 20.18. Machine Learning Algorithms: Popular Algorithms for Data Science and Machine Learning, 2nd Edition. 2nd ed. Birmingham : Packt Publishing
Accession Number: 2106304	Brett Lantz. 2019. Machine Learning with R : Expert Techniques for Predictive Modeling. Third edition. Birmingham, UK : Packt Publishing.
Publication Paper:	
DOI: 10.1109/ICMLA52953.2021.00197	Agnes Tegen; Paul Davidsson; Jan A. Persson. 2021. Active Learning and Machine Teaching for Online Learning: A Study of Attention and Labelling Cost. Published in: 2021 20th IEEE International Conference on Machine Learning and Applications (ICMLA). Retrieved from https://ieeexplore.ieee.org/document/9680069.
DOI: 10.1109/MLSP55214.2022.9943373	Goutam Kumar Sahoo; Keerthana Kanike; Santos Kumar Das; Poonam Singh. 2022. Machine Learning-Based Heart Disease Prediction: A Study for Home Personalized Care. Published in: 2022 IEEE 32nd International Workshop on Machine Learning for Signal Processing (MLSP). Retrieved from https://ieeexplore.ieee.org/document/9943373

Prepared: **Computer Science Department**
1st Semester 2023-2024

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De La Salle University - Dasmariñas

COLLEGE OF SCIENCE AND COMPUTER STUDIES
COMPUTER SCIENCE DEPARTMENT

PROGRAM LEARNING OUTCOME - COURSE LEARNING OUTCOME MAPPING TABLE FOR BS COMPUTER SCIENCE

COURSE CODE : S-CSIS311
COURSE TITLE : INTRODUCTION TO MACHINE LEARNING
COURSE TYPE : Lecture
COURSE CREDIT : 2 Units

Course Learning Outcome	Program Learning Outcome									
	PLO 1	PLO 2	PLO 3	PLO 4	PLO 5	PLO 6	PLO 7	PLO 8	PLO 9	PLO 10
1. Determine a wide variety of learning algorithms.	/	/	/		/					
2. Demonstrate how to evaluate models generated from data.		/	/	/	/		/			
3. Develop an appreciation for what is involved in learning models from data.		/	/		/		/	/		

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4. Apply the algorithm to a real-world problem	/	/	/	/	/	/	/	/	/	/
5. Formulate the optimize models learned and report on the expected accuracy that can be achieved by applying the models.				/	/		/	/	/	/

NOTE: Provide a check mark on the areas in which the program learning outcome (PLO) is hit by the course learning outcome (CLO)

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COLLEGE OF SCIENCE AND COMPUTER STUDIES COMPUTER SCIENCE DEPARTMENT

COMMON GE LEARNING OUTCOME - COURSE LEARNING OUTCOME MAPPING TABLE FOR BS COMPUTER SCIENCE

COURSE CODE : S-CSIS311

COURSE TITLE : INTRODUCTION TO MACHINE LEARNING

COURSE TYPE : Lecture

COURSE CREDIT : 2 Units

Course Learning Outcome	Common GE Learning Outcome				
	CGE1	CGE2	CGE3	CGE4	CGE5
1. Determine a wide variety of learning algorithms.	/			/	/
2. Demonstrate how to evaluate data driven models.	/			/	/
3. Develop an appreciation for what is involved in learning models from data.				/	/
4. Apply the algorithm to a real-world problem			/	/	/
5. Formulate the optimize models learned and report on the expected accuracy that can be achieved by applying the models	/		/	/	/

NOTE: Provide a check mark on the areas in which the program learning outcome (PLO) is hit by the course learning outcome (CLO)

COMMON GE LEARNING OUTCOME

1. Critiques/evaluates individual's character, actions, and decisions based on Christian-like perspective
2. Assesses situations and proposes ways that benefit the country, particularly its environment and its marginalized sector
3. Explains in various occasions the relationship of the course/subject with the other subjects within or outside the discipline
4. Recommends ways to improve the subject/course

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GRADUATE ATTRIBUTES (DESCRIPTORS/INSTITUTIONAL LEARNING OUTCOMES) – PROGRAM LEARNING OUTCOME MAPPING TABLE FOR BS COMPUTER SCIENCE

Program Learning Outcome	Graduate Attributes/Institutional Learning Outcomes				
	GA1	GA2	GA3	GA4	GA5
1. Apply concepts of computing and mathematics in different domains of computer science.				/	/
2. Identify and analyze problems and define computing requirements appropriate to their solutions.					/
3. Design, develop, test and evaluate various solutions using algorithms for complex computing problems that take full advantage of usability of computer resources.			/	/	/
4. Document computer-based systems by writing effective reports for risk- assessment impact.				/	/
5. Work effectively as an individual or as a member of a team during software development.			/	/	/
6. Communicate effectively with the computing community and non-computer science professionals about computing activities.				/	/
7. Recognize the legal, social, ethical and professional issues involved in the utilization of computer technology and address these issues.	/	/	/		
8. Engage in continuing professional development through performing independent or collaborative research as computing professionals.			/	/	/

NOTE: Provide a check mark on the areas in which the Graduate Attribute (Descriptors/Institutional Learning Outcome) is hit by the program learning outcome (PLO). Kindly refer to the descriptors (institutional learning outcomes) to clearly understand what each attribute refers to or expect from its graduates.

Graduate Attribute (GA)	**De criptors (Institutional Learning Outcome)[1]**
1. God-centered	a) Creates an environment where the experience of God is lived and shared b) Practices honesty, fairness, truth and integrity in all aspects of life (personal and professional lives) c) Observes and maintains ethical standards in dealing with the different stakeholders d) Integrates Christian perspectives and values in all undertakings e) Manifests humility and respect in relating with other people
2. Patriotic	a) Manifests a deep sense of nationalism by integrating history, arts, and culture in their daily lives b) Participates responsibly and collaboratively in the discussion and resolution of issues within local, national, and international contexts c) Engages actively in political, social, economic, and cultural transformation for nation building d) Brings pride and honor to the community and the country e) Patronizes locally produced products and promotes them globally
3. Socially Responsive	a) Demonstrates concern and compassion for the plight of the vulnerable and marginalized sectors of society by participating meaningfully in the process of social transformation b) Engages in ecological advocacies as responsible stewards of God's creation c) Continuously works in solidarity with people and institutions to effect liberating action d) Brings forth awareness and deeper understanding of social realities e) Manifests spirit of volunteerism through social actions that improve communities
4. Globally Competitive	a) Takes progressive responsibility for own learning and development b) Promotes critical and creative thinking, self-knowledge and self-mastery c) Is committed to rise above mediocrity d) Shows dedication, commitment and love for work to attain exceptional results e) Exhibits innovativeness and creativity in various contexts
5. Professionally Competent	a) Possesses leadership qualities of **C**ompetence, **H**onesty, **A**ccountability, **R**esponsibility and **T**ransparency (CHART) b) Effectively communicates in various forms of media c) Strives to be at the top of their chosen fields d) Effectively works and collaborates in a multi-disciplinary team e) Adapts to changes in the workplace, local society and global community